Revisions

Rev.	Date	Author	Description
A	23/12/2021	R.F	Creation nouvelle FBA + claire en refonte pour Tyva/Sevcon
B	04/03/2025	R.F	Ajout d'une resistance 120 ohm sur circuit can drive
C	24/03/2025	R.F	Ajout des rallonges

Fourreaux

N°1

Emergency brake
Controler Left
Controller pump
Charger
Resistor 120 ohms

N°2

Emergency brake
Controller right
Battery 350 V

N°3

MC42 cmd
24V converter

	1	2	3	4	5	6	7	8	9	10	11	12
A	30 Batt 86 / 85 87a 87			Bat ignti. (3A)		Charger (20A)		Bat logic (3A)		Brake (20A)		
B										Input DCDC (20A)		
C	30 Var 86 / 85 87a 87			Var 1 (20A)		Var 2 (20A)		Var 3 (20A)				
D												
E												

FuseBox_FBA

Cavity	To	Conductor	Color	Size	Notes
A1	Battery_400V_Switched_Supply.A	W1.Red		1,5 mm²	
A3	MC42_Cmd.B	W55.Orange		1 mm²	
A4	FuseBox_FBA.B3	W5.Red		1,5 mm²	
A5	Battery_350V.E	W8.Blue		1 mm²	Battery_Fuse_20A
A6	Splice_12V	W6.Red		1,5 mm²	Charger_Fuse_20A
A7	Charger.A	W88.Red		1,5 mm²	Charger_Fuse_20A
A8	Battery_400V_Switched_Supply.B	W4.Red		1,5 mm²	
A9	Battery_350V.A	W32.Orange		1,5 mm²	
A10	Splice_12V	W90.Red		1,5 mm²	
A11	Splice_Brake	W92.Red		1,5 mm²	
B1	MC42_Cmd.C	W15.Violet		1 mm²	Batt_Relay_86
B3	FuseBox_FBA.A4	W5.Red		1,5 mm²	
B10	24V_Converter_10807.A	W37.Gray		1,5 mm²	
B11	Splice_12V_Controller	W38.Gray		1,5 mm²	
C1	Splice_12V	W2.Red		1,5 mm²	Cont_Relay_30
C3	MC42_Cmd.D	W56.Orange		1 mm²	
C4	Splice_24V_Plus	W30.Green		1 mm²	Controler_1_Fuse_20A
C5	Controler_Right.A	W47.Green		1 mm²	Controler_1_Fuse_20A
C6	Splice_24V_Plus	W36.Green		1 mm²	Controler_2_Fuse_20A
C7	Controler_Left.A	W46.Green		1 mm²	Controler_2_Fuse_20A
C8	Splice_12V_Controller	W27.Gray		1,5 mm²	Controler_3_Fuse_20A
C9	Controler_Pump.A	W91.Red		1,5 mm²	Controler_3_Fuse_20A
D1	MC42_Cmd.A	W17.Green		1 mm²	
D3	Splice_12V_Controller	W28.Gray		1,5 mm²	Cont_relay_87

Splice_12V_Controller

From	To	Conductor	Color	Size
Splice_12V_Controller	FuseBox_FBA.B11	W38.Gray		1,5 mm²
Splice_12V_Controller	FuseBox_FBA.C8	W27.Gray		1,5 mm²
Splice_12V_Controller	FuseBox_FBA.D3	W28.Gray		1,5 mm²

FuseBox_FBA [1, 14, 16, 25, 43, 44] — #22105 — [17] — 1 — Splice_12V_Controller [27]

Splice_Brake

From	To	Conductor	Color	Size
Splice_Brake	Brake_Emergency_Left.A	W93.Red		1,5 mm²
Splice_Brake	Brake_Emergency_Right.A	W108.Red		1,5 mm²
Splice_Brake	FuseBox_FBA.A11	W92.Red		1,5 mm²

1 — Splice_Brake [27]
1 m — [17] — Emergency Brake — Brake_Emergency_Left [7, 14, 19, 43] — Bouchon2 [10, 19, 44]
1 m — [17] — Emergency Brake — Brake_Emergency_Right [7, 14, 19, 43] — Bouchon1 [10, 19, 44]

Brake_Emergency_Left

Cavity	To	Conductor	Color	Size
A	Splice_Brake	W93.Red		1,5 mm²
B	Splice_0V	W26.Black		1,5 mm²

Brake_Emergency_Right

Cavity	To	Conductor	Color	Size
A	Splice_Brake	W108.Red		1,5 mm²
B	Splice_0V	W83.Black		1,5 mm²

Splice_0V

From	To	Conductor	Color	Size
Splice_0V	24V_Converter_10807.B	W79.Black		1,5 mm²
Splice_0V	Battery_350V.B	W78.Black		1,5 mm²
Splice_0V	Battery_350V.D	W81.Black		1,5 mm²
Splice_0V	Battery_350V.H	W54.Black		1 mm²
Splice_0V	Brake_Emergency_Left.B	W26.Black		1,5 mm²
Splice_0V	Brake_Emergency_Right.B	W83.Black		1,5 mm²
Splice_0V	Charger.B	W84.Black		1,5 mm²
Splice_0V	Controler_Pump.B	W9.Black		1 mm²
Splice_0V	Supply.B	W52.Black		4 mm²

2 — Splice_0V [27] / Splice_12V [27]
0,4 m — [17] — Supply (FBA02)(PS05) — Supply [3, 11, 20, 45]

Supply

Cavity	To	Conductor	Color	Size
A	Splice_12V	W51.Red		4 mm²
B	Splice_0V	W52.Black		4 mm²

Splice_12V

From	To	Conductor	Color	Size
Splice_12V	FuseBox_FBA.A6	W6.Red		1,5 mm²
Splice_12V	FuseBox_FBA.A10	W90.Red		1,5 mm²
Splice_12V	FuseBox_FBA.C1	W2.Red		1,5 mm²
Splice_12V	Supply.A	W51.Red		4 mm²

1 m — [17] — Battery 400V Supply 12V (FBA12)(PS02) — Battery_400V_Switched_Supply [9, 15, 24, 43]

Battery_400V_Switched_Supply

Cavity	To	Conductor	Color	Size
A	FuseBox_FBA.A1	W1.Red		1,5 mm²
B	FuseBox_FBA.A8	W4.Red		1,5 mm²

Splice_Can_Low

From	To	Conductor	Color	Size
Splice_Can_Low	Battery_350V.G	W3.Blue		1 mm²
Splice_Can_Low	CanBus_Drive_From_MC42.B	W106.Blue		1 mm²
Splice_Can_Low	Charger.D	W102.Blue		1 mm²
Splice_Can_Low	Controler_Left.D	W103.Blue		1 mm²
Splice_Can_Low	Controler_Right.D	W105.Blue		1 mm²
Splice_Can_Low	KeyBox.B	W18.Blue		1 mm²
Splice_Can_Low	resistor_120_ohm_can_drive_F.B	W10.Blue		0,75 mm²

0,8 m — [17] — MC42 (FBA_01)(MC_13) — MC42_Cmd [4, 12, 21, 46]

MC42_Cmd

Cavity	To	Conductor	Color	Size
A	FuseBox_FBA.D1	W17.Green		1 mm²
B	FuseBox_FBA.A3	W55.Orange		1 mm²
C	FuseBox_FBA.B1	W15.Violet		1 mm²
D	FuseBox_FBA.C3	W56.Orange		1 mm²

2 — Splice_Can_Low [27] / Splice_Can_High [27]
1 m — [17] — Can Drive (FB_03)(MC_20) — CanBus_Drive_From_MC42 [2, 13, 22, 46]

CanBus_Drive_From_MC42

Cavity	To	Conductor	Color	Size
A	Splice_Can_High	W94.White		1 mm²
B	Splice_Can_Low	W106.Blue		1 mm²

Splice_Can_High

From	To	Conductor	Color	Size
Splice_Can_High	Battery_350V.F	W29.White		1 mm²
Splice_Can_High	CanBus_Drive_From_MC42.A	W94.White		1 mm²
Splice_Can_High	Charger.C	W96.White		1 mm²
Splice_Can_High	Controler_Left.C	W97.White		1 mm²
Splice_Can_High	Controler_Right.C	W99.White		1 mm²
Splice_Can_High	KeyBox.A	W21.White		1 mm²
Splice_Can_High	resistor_120_ohm_can_drive_F.A	W20.White		0,75 mm²

resistor_120_ohm_can_drive_F [2, 13, 22, 48] — resistor_120_ohm_can_drive_M [6, 12, 22, 48] — 1 — 120_Ohm1 [26]

resistor_120_ohm_can_drive_F

Cavity	To	Conductor	Color	Size
A	Splice_Can_High	W20.White		0,75 mm²
B	Splice_Can_Low	W10.Blue		0,75 mm²

0,8 m — [17] — Charg (FBA_04)(CB_01) — Charger [5, 13, 18, 46]

Charger

Cavity	To	Conductor	Color	Size
A	FuseBox_FBA.A7	W88.Red		1,5 mm²
B	Splice_0V	W84.Black		1,5 mm²
C	Splice_Can_High	W96.White		1 mm²
D	Splice_Can_Low	W102.Blue		1 mm²

0,8 m — [17] — Battery (xFBA_05)(TY01) — Battery_350V [8, 13, 23, 47, 48]

Battery_350V

Cavity	To	Conductor	Color	Size
A	FuseBox_FBA.A9	W32.Orange		1,5 mm²
B	Splice_0V	W78.Black		1,5 mm²
D	Splice_0V	W81.Black		1,5 mm²
E	FuseBox_FBA.A5	W8.Blue		1 mm²
F	Splice_Can_High	W29.White		1 mm²
G	Splice_Can_Low	W3.Blue		1 mm²
H	Splice_0V	W54.Black		1 mm²

Splice_Can_Blue

From	To	Conductor	Color	Size
Splice_Can_Blue	CanBus_J1939_From_MC42.B	W7.Blue		1 mm²
Splice_Can_Blue	Controler_Pump.D	W22.Blue		1 mm²
Splice_Can_Blue	Resistor_120_Ohm_J1939_F.B	W12.Blue		1 mm²

0,4 m — [17] — Resistor 120 Ohm — Resistor_120_Ohm_J1939_F [2, 13, 22, 46] — 1 — 120_Ohm [26] — Resistor_120_Ohm_J1939_M [6, 12, 22, 46]

Resistor_120_Ohm_J1939_F

Cavity	To	Conductor	Color	Size
A	Splice_Can_white	W11.White		1 mm²
B	Splice_Can_Blue	W12.Blue		1 mm²

2 — Splice_Can_Blue [27] / Splice_Can_white [27]
1 m — [17] — J1939 (FBA06)(MC19) — CanBus_J1939_From_MC42 [2, 13, 22, 46]

Splice_Can_white

From	To	Conductor	Color	Size
Splice_Can_white	CanBus_J1939_From_MC42.A	W19.White		1 mm²
Splice_Can_white	Controler_Pump.C	W16.White		1 mm²
Splice_Can_white	Resistor_120_Ohm_J1939_F.A	W11.White		1 mm²

CanBus_J1939_From_MC42

Cavity	To	Conductor	Color	Size
A	Splice_Can_white	W19.White		1 mm²
B	Splice_Can_Blue	W7.Blue		1 mm²

0,8 m — [17] — PUMP (FBA_11)(HV_01) — Controler_Pump [5, 13, 18, 46]

Controler_Pump

Cavity	To	Conductor	Color	Size
A	FuseBox_FBA.C9	W91.Red		1,5 mm²
B	Splice_0V	W9.Black		1 mm²
C	Splice_Can_white	W16.White		1 mm²
D	Splice_Can_Blue	W22.Blue		1 mm²

0,8 m — [17] — Right (FBA_07)(SV_02) — Controler_Right [5, 13, 18, 46]

Controler_Right

Cavity	To	Conductor	Color	Size
A	FuseBox_FBA.C5	W47.Green		1 mm²
B	Splice_24V_Minus	W49.Black		1 mm²
C	Splice_Can_High	W99.White		1 mm²
D	Splice_Can_Low	W105.Blue		1 mm²

0,8 m — [17] — Left (FBA_08)(SV_01) — Controler_Left [5, 13, 18, 46]

Controler_Left

Cavity	To	Conductor	Color	Size
A	FuseBox_FBA.C7	W46.Green		1 mm²
B	Splice_24V_Minus	W48.Black		1 mm²
C	Splice_Can_High	W97.White		1 mm²
D	Splice_Can_Low	W103.Blue		1 mm²

Splice_24V_Plus

From	To	Conductor	Color	Size
Splice_24V_Plus	24V_Converter_10807.C	W45.Green		1 mm²
Splice_24V_Plus	FuseBox_FBA.C4	W30.Green		1 mm²
Splice_24V_Plus	FuseBox_FBA.C6	W36.Green		1 mm²

2 — Splice_24V_Plus [27] / Splice_24V_Minus [27]
0,5 m — [17] — Canbus keybox (FBA_10)(KE01) — KeyBox [2, 13, 22, 46]

KeyBox

Cavity	To	Conductor	Color	Size
A	Splice_Can_High	W21.White		1 mm²
B	Splice_Can_Low	W18.Blue		1 mm²

Splice_24V_Minus

From	To	Conductor	Color	Size
Splice_24V_Minus	24V_Converter_10807.D	W23.Black		1 mm²
Splice_24V_Minus	Controler_Left.B	W48.Black		1 mm²
Splice_24V_Minus	Controler_Right.B	W49.Black		1 mm²

0,8 m — [17] — 24V (FBA_12) — 24V_Converter_10807 [5, 13, 18, 46]

24V_Converter_10807

Cavity	To	Conductor	Color	Size
A	FuseBox_FBA.B10	W37.Gray		1,5 mm²
B	Splice_0V	W79.Black		1,5 mm²
C	Splice_24V_Plus	W45.Green		1 mm²
D	Splice_24V_Minus	W23.Black		1 mm²

Rallonges

Pump_M

Cavity	To	Conductor	Color	Size
A	Pump_F.A	W39.Red		1 mm²
B	Pump_F.B	W40.Black		1 mm²
C	Pump_F.C	W41.White		0,75 mm²
D	Pump_F.D	W42.Black		0,75 mm²

Pump_M [4] — 2,5 m — Pump_F [5]

Pump_F

Cavity	To	Conductor	Color	Size
A	Pump_M.A	W39.Red		1 mm²
B	Pump_M.B	W40.Black		1 mm²
C	Pump_M.C	W41.White		0,75 mm²
D	Pump_M.D	W42.Black		0,75 mm²

Right_M

Cavity	To	Conductor	Color	Size
A	Right_F.A	W31.Red		1 mm²
B	Right_F.B	W33.Black		1 mm²
C	Right_F.C	W34.White		0,75 mm²
D	Right_F.D	W35.Black		0,75 mm²

Right_M [4] — 3,5 m — Right_F [5]

Right_F

Cavity	To	Conductor	Color	Size
A	Right_M.A	W31.Red		1 mm²
B	Right_M.B	W33.Black		1 mm²
C	Right_M.C	W34.White		0,75 mm²
D	Right_M.D	W35.Black		0,75 mm²

Left_M

Cavity	To	Conductor	Color	Size
A	Left_F.A	W43.Red		1 mm²
B	Left_F.B	W44.Black		1 mm²
C	Left_F.C	W50.White		0,75 mm²
D	Left_F.D	W53.Black		0,75 mm²

Left_M [4] — 2,5 m — Left_F [5]

Left_F

Cavity	To	Conductor	Color	Size
A	Left_M.A	W43.Red		1 mm²
B	Left_M.B	W44.Black		1 mm²
C	Left_M.C	W50.White		0,75 mm²
D	Left_M.D	W53.Black		0,75 mm²

Title: #36098 - FuseBox Electric Drive Sevcon/Tyva X100/Sport (FBA) - IND B
Description
Revision: ----

	Name	Date
Drawn By	----	----
Eng Approved	----	----
Mfg Approved	----	----
QA	----	----

This document contains confidential and proprietary information that cannot be reproduced or divulged, in whole or in part, without written authorization.

Title:	#36098 - FuseBox Electric Drive Sevcon/Tyva X100/Sport (FBA) - IND B
Description:	

Id	Type	Part Number	Quantity	Description	Manufacturer
1	Connector	#26535 - Littelfuse_280_60_way_FuseBox	1	Littelfuse PDM21001LXM	Littelfuse
2	Connector	#14869 - Delphi_150_2_way_female	5	Delphi 12052641	Delphi
3	Connector	#17183 - Delphi_480_2_way_male	1	Delphi 12065863	Delphi
4	Connector	#26525 - Delphi_150_4_way_male	4	Delphi 12160825	Delphi
5	Connector	#28050 - Delphi_150_4_way_female	8	Delphi 12160482-B	Delphi
6	Connector	#15215 - Delphi_150_2_way_male	2	Delphi 12162000	Delphi
7	Connector	#17351 - Delphi_280_2_way_female	2	Delphi 15300027	Delphi
8	Connector	#13000 - Delphi_150_8_way_female	1	Delphi 12047937	Delphi
9	Connector	Delphi_280_3_way_male	1	#14168 - 829-15300003	Delphi
10	Connector	#16777 - Delphi_280_2_way_male	2	Delphi 15300002	Delphi
11	Contact	#17237 Delphi_480_contact_male 5mm² 3.49-5.24mm	2	Delphi 12052172	Delphi
12	Contact	#10019 Delphi_150_contact_male 0.5-1.0mm² 1.69-2.78mm	8	Delphi 12045773	Delphi
13	Contact	#13657 Delphi_150_contact_female 0.5-1.0mm² 2.03-2.80mm	38	Delphi 12048074	Delphi
14	Contact	#14173 Delphi_280_contact_female 2.0-3.0mm² 2.45-3.80mm	30	Delphi 12129493	Delphi
15	Contact	#13581 Delphi_280_contact_male 0.5-0.8mm² 1.69-2.78mm	3	Delphi 12048159	Delphi
16	Fuse	#20949 Fuse 20A Mini	8	Littelfuse 0297020.WXNV	Littelfuse
17	Label	Generic White Label	16	Generic Label	
18	Lock	#22024 - Delphi_150_4_way_female_lock	5	Delphi 12160480	Delphi
19	Lock	#18797 - Delphi_280_2_way_lock	4	Delphi 15300014-B	Delphi
20	Lock	#10273 - Delphi_480_2_way_lock	1	Delphi 12059897	Delphi
21	Lock	#23010 - Delphi_150_4_way_male_lock	1	Delphi 12176368	Delphi
22	Lock	#19468 - Delphi_150_2_way_lock	7	Delphi 12052634	Delphi
23	Lock	#12834 - Delphi_150_8_way_female_lock	1	Delphi 15324525	Delphi
24	Lock	#18566 - Delphi_280_3_Way_lock	1		
25	Relay	#36314 5 PIN SPDT sealed relay w/ suppress.diode Micro 280	2	Song Chuan 3011CSD112VDC	Song Chuan
26	Resistor	#27646 120Ω ±1%, 0.6W metal film resistor	2	TE Connectivity LR1F120R	TE Connectivity
27	Splice	Generic Splice	10	Generic Splice	
28	Wire	1 mm² - Orange	≥ 1,6 m		
29	Wire	1.5 mm² Black	≥ 1 m	Generic Wire	
30	Wire	1.5mm² - Black	≥ 4,2 m		
31	Wire	1.5mm² - Red	≥ 5,6 m		
32	Wire	1mm² - Blue	≥ 7,7 m		
33	Wire	1mm² - White	≥ 6,9 m		
34	Wire	1mm² Black	≥ 12,5 m		
35	Wire	4 mm² - Black	≥ 0,4 m		
36	Wire	4 mm² - Red	≥ 0,4 m		
37	Wire	1.5mm² - Gray	≥ 0,8 m		
38	Wire	1 mm² Purple	≥ 0,8 m		
39	Wire	1mm² Green	≥ 3,2 m		
40	Wire	1.5 mm² - Orange	≥ 0,8 m		
41	Wire	0,75² Blue			
42	Wire	0,75² White	≥ 5 m		
43	Wire Seal	#13381 - Delphi 280 Cable seal – Green– 2.03-2.85	33	Delphi 12015284	Delphi
44	Wire Seal	#19508 - Delphi 280 Cavity plug	38	Delphi 12010300	Delphi
45	Wire Seal	#15414 - Delphi 480 Cable seal – Red – 2.81-3.76	2	Delphi 12052452	Delphi
46	Wire Seal	#17898 - Delphi 150 Cable seal – White – 1.60-2.15	34	Delphi 12089442	Delphi
47	Wire Seal	#18901 - Delphi 150 Cable seal – Dark-red – 2.03-2.85	5	Delphi 12052924	Delphi
48	Wire Seal	#14820 - Delphi 150 Cable seal – Blue – 1.29-1.70	7	Delphi 12052925	Delphi
49	Wire	1mm² Red	8,5 m		
50	Wire	0.75² White	3,5 m		
51	Wire	0.75² Black	8,5 m		